

Preliminary Data Analysis

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CoNVO statement

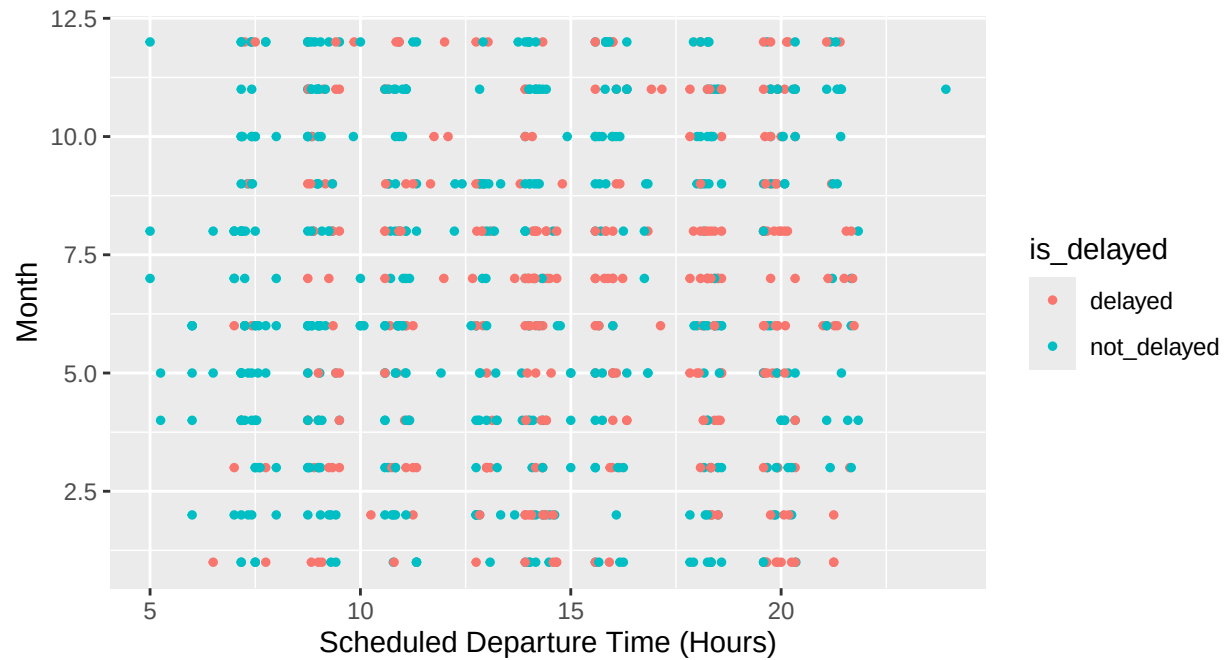
United Airlines operates thousands of flights every day across a large and complex network, and identifying flights at high risk of delay months in advance is a key challenge for the network planning team. Having better visibility into which flights are likely to be delayed is an important step toward building more reliable and operationally efficient schedules.

Understanding what drives flight delays will improve foresight into where and when delays are most likely to occur. Certain routes, markets, and time slots are consistently more prone to delays than others. This will help highlight routes and markets that are consistently delayed, giving planning teams a clearer picture of where the network is underperforming. With this information, better long-term scheduling decisions can be made that improve on-time performance and reduce delays across the network.

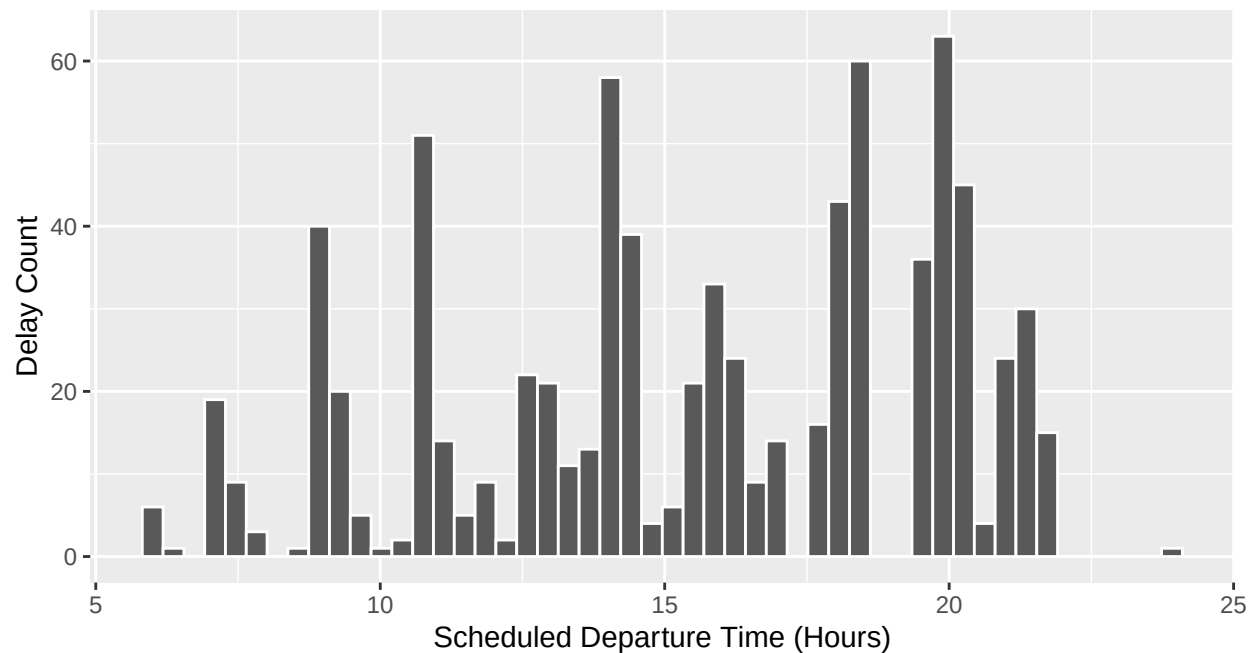
Using historical flight data, we will build a model that predicts the probability of departure delays for flights across the network. The dataset includes flight-level information on scheduled times, routes, and day of week, providing a strong foundation for identifying where delays are most likely to occur. By identifying the key factors that contribute to delays, the model will give network planning teams the information they need to adjust routes and time slots where schedule changes could improve reliability and operational performance. Flight-level predictions will allow planners to move beyond aggregate on-time statistics and evaluate specific flights and markets that may benefit from schedule adjustments. The model will give planners a way to simulate the impact of schedule changes before they are made, helping the team make more informed decisions.

Preliminary results for departures out of ORD (Chicago O'Hare International Airport) will be presented to a VP in Network Planning. ORD is a United hub while being one of the busiest and most delay-prone airports in the country, making it a natural starting point for this analysis. If successful, the model will be run across all flights in the network on an ongoing basis, giving planning teams regular insights into where delays are expected and where schedules may need to be adjusted. The long term goal is to make this model a regular part of the network planning process to help the team build better schedules and reduce delays across the network.

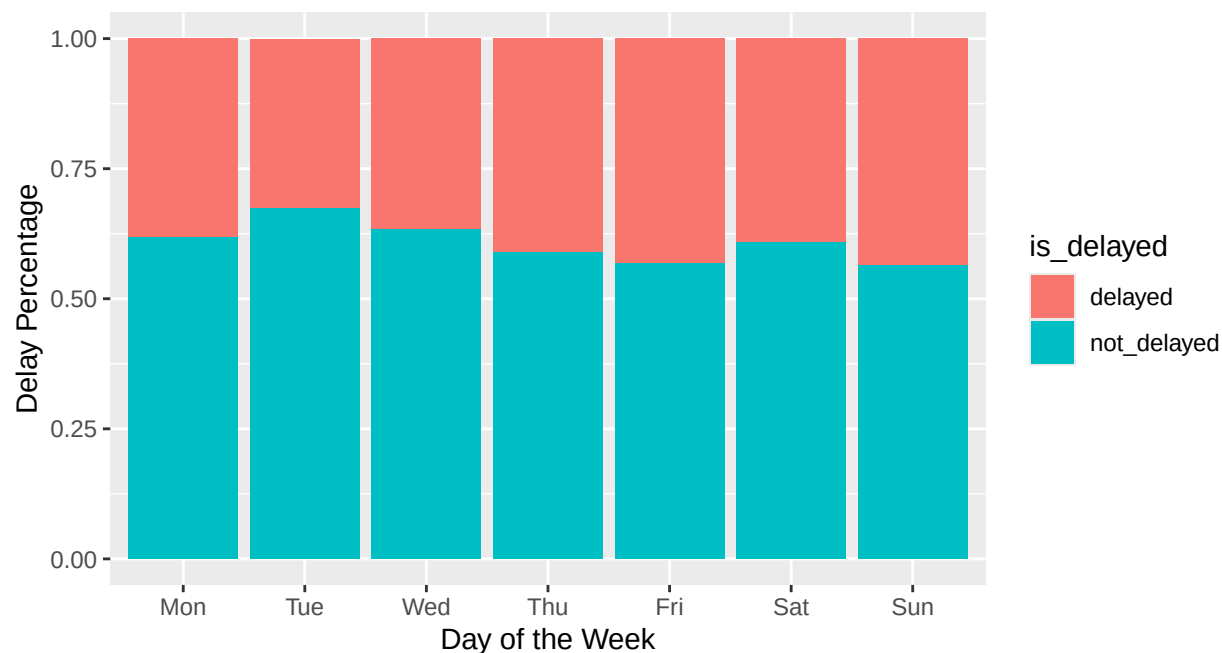
Data Visualization



The plot above shows how delayed flights are dispersed throughout the day in different months at ORD. There tend to be more delayed flights in the afternoon compared to the morning during all seasons, and there are certain seasons that have more delays than others.



The histogram above shows counts of delays through the day at ORD. Similar to the last plot, this graph shows more delays later in the day as well as the bank structure (peak times which increase the amount of connections United is able to offer) of the United schedule at ORD.



The plot above shows how the percentage of flights delayed changes by the day of the week. At ORD, Tuesday has the highest percentage of on-time flights followed by Wednesday, Monday, Saturday, Thursday, Friday, and Sunday.

Model Fitting

Below is the proportion of each level of the outcome. Models will be fit with a departure airport of ORD.

```
##
##      delayed not_delayed
## 0.3924153  0.6075847
```

Logistic Regression

```
##   parameter      ROC      Sens      Spec      ROCSD      SensSD      SpecSD
## 1      none 0.6734765 0.4149873 0.8009426 0.002872104 0.001994414 0.002875889
```

Random forest

```
##   mtry      ROC      Sens      Spec      ROCSD      SensSD      SpecSD
## 2     2 0.6934875 0.3675154 0.8679604 0.003169224 0.009252571 0.006613114
```

Boosting

```
##   shrinkage interaction.depth n.minobsinnode n.trees      ROC      Sens
## 9      0.1              3          10      150 0.6999923 0.396721
##      Spec      ROCSD      SensSD      SpecSD
## 9 0.8525179 0.003972014 0.005105384 0.003027434
```

Model comparison

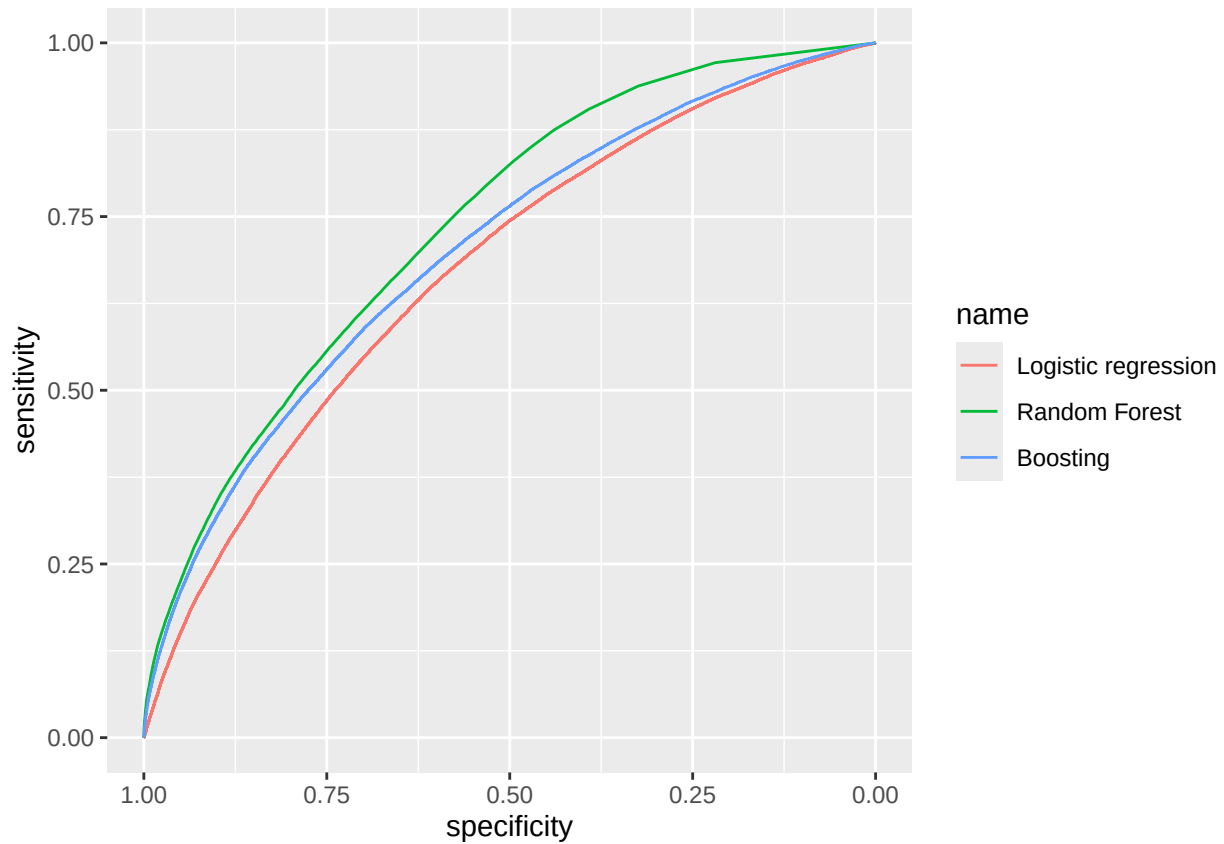


Table 1: Model Comparison

Model	CV_ROC	AUC	Sensitivity	Specificity
Logistic Regression	0.673	0.674	0.415	0.801
Random Forest	0.693	0.736	0.368	0.868
Boosting	0.700	0.701	0.397	0.853

As seen in the ROC curve above, the Random Forest model has the best AUC followed by the Boosting model and then the Logistic Regression model. When looking at the cross-validated AUC (notated as CV_ROC in the table), it is clear that the Random Forest model is over fit as it performed better on the training data than the CV test data. These models result in a similar level of AUC with all three models have an AUC that indicates improvement over selecting the most common outcome.

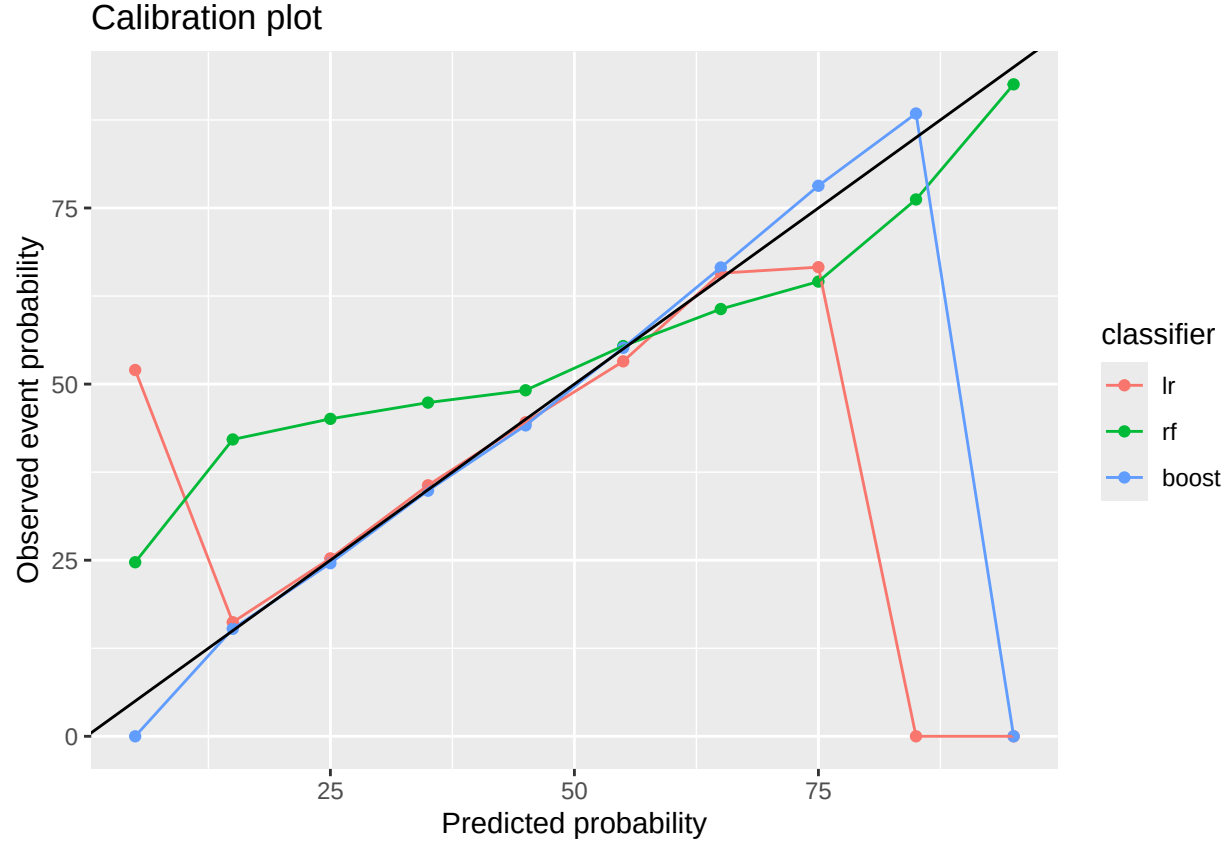


Table 2: Calibration Summary by Model

Bin	Logistic Regression		Random Forest		Boosting	
	LR Avg Percent	LR Count	RF Avg Percent	RF Count	Boost Avg Percent	Boost Count
5	52.000	13	24.710	22811	0.000	0
15	16.196	1984	42.140	6635	15.260	2809
25	25.259	11503	45.067	5271	24.576	9538
35	35.630	13857	47.371	4127	34.866	15499
45	44.576	12965	49.124	4205	44.138	13614
55	53.228	17765	55.399	4689	55.155	11678
65	65.752	10515	60.655	4850	66.554	8861
75	66.598	323	64.555	5522	78.155	5438
85	0.000	0	76.213	8279	88.414	1488
95	0.000	0	92.555	2536	0.000	0

As seen in the calibration plot and table above, the Logistic Regression and Boosting models are calibrated well. The deviation from the line in the plot by the Boosting model is caused by 0 predictions made in either of the [0:0.1] or [0.9:1] bins. A similar pattern is seen with the Logistic Regression model, though the entire distribution is biased more towards the more common outcome (no delay). When looking at the Random Forest calibration plot and table, the model over predicts low likelihood outcomes and under predicts high likelihood outcomes, indicating it is very biased towards the no delay outcome.

Plan of work

Going forward, I want to incorporate more predictors into the model. I plan to create rolling statistics such as scheduled flights in the last 30 minutes for both United and all airlines at ORD. I will also include the arrival airport as a predictor as it was excluded in these models. Ideally, these new predictors will improve the quality of the model predictions and result in better decisions on a flight level. I also want to compare how models will perform when they are fit on non-hub departures such as flights leaving PIT.